

Ricardo (Zhicong) Xie

Building Services / HVAC Intern | Revit MEP, MVHR/ERV, CAD and Commissioning

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LinkedIn: <https://www.linkedin.com/in/ricardo-xie-6394643b4> | Portfolio: <https://zxie588-alt.github.io> | GitHub: <https://github.com/zxie588-alt>

NZ student visa | Full-time eligible over scheduled summer break 17 Nov 2026 - late Feb 2027 | No employer visa support needed for this internship | Open NZ-wide

PROFILE

- Auckland-based mechanical master's student with a NZ residential MVHR/ERV Revit MEP project.
- Built layout, airflow schedule, duct velocity checks, pressure estimate, equipment note and commissioning plan.
- Project screenshots and source links: <https://zxie588-alt.github.io>

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Study focus and portfolio work in mechanical, building-services, manufacturing and analysis contexts.

Study record/focus: Completed: Computational Fluid Dynamics, Engineering Methods, Composite Materials. Registered S2: Building Services, Smart Manufacturing, Manufacturing and Industrial Systems.

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PRACTICAL AND TEAM EXPERIENCE

- Engineering team leadership: led task scoping, workload split and final technical presentation in team design work.
- Workshop practice: prepared AutoCAD/DXF laser-cut geometry and checked cut parts against drawings.
- Hands-on exposure: CNC machining, workholding, cutting fluid, chip formation and 3D printing workflow.
- Lab practice: interpreted force-displacement tests and controlled a robotic arm in training sessions.
- Technical communication: former student news lead; strong project records, photography and presentation visuals.

PROJECT WORK

Small Residential MVHR / ERV Revit MEP Coordination

Revit 2026 MEP coordination, MVHR/ERV layout, airflow schedule, SolidWorks, AutoCAD/DXF, NZBC G4/H1 context, commissioning notes

- Built a small NZ residential MVHR/ERV coordination package with Revit MEP, SolidWorks and AutoCAD/DXF evidence.
- Created plan, 3D view, airflow schedule, sheet, review comments and simple clash/coordination notes.
- Documented preliminary airflow, pressure-loss, balancing and commissioning actions for the next design iteration.

Thermal-Fluid Simulation & Verification - Solar Chimney Case Study

ANSYS Fluent/CFX, conjugate heat transfer, natural convection

- Built a CFD workflow for a laboratory-scale solar chimney with coupled air, absorber and cover domains.
- Assessed model credibility through mesh sensitivity, mass/heat balance and Reynolds/Rayleigh scaling.
- Documented model boundaries, limitations and recommended physical data points needed for future experimental correlation.

Manufacturing Workshop Training & DXF Preparation

Laser cutting, AutoCAD/DXF, CNC machining exposure, 3D printing workflow

- Prepared AutoCAD/DXF geometry for laser cutting and checked the cut plate against drawing intent.
- Practised CNC workflow: workholding, tool motion, cutting fluid, chip formation and shop safety.
- Reviewed machined gears and shafts for tolerance, edge quality and tool-access constraints.

Engineering Practice and Automotive Structure Training

Mechanics testing, robotic arm control, automotive structure walkthrough

- Worked with mechanics testing equipment and interpreted force-displacement results from a physical test setup.
- Practised robotic arm control and basic automation interaction in a lab setting.
- Led a small automotive-structure walkthrough covering seat/interior systems and vehicle context.

TECHNICAL SKILLS

- Revit MEP: MVHR/ERV layout, coordination plan, 3D view, airflow schedule and sheet setup.
- HVAC checks: airflow schedule, duct velocity, pressure-loss estimate and fan-duty assumptions.
- CAD support: SolidWorks, AutoCAD/DXF, component lists and colour-coded coordination layouts.
- NZ context: NZBC G4/H1 awareness, Healthy Homes ventilation and commissioning notes.
- Analysis support: heat transfer, natural-convection CFD and sensor testing workflow.